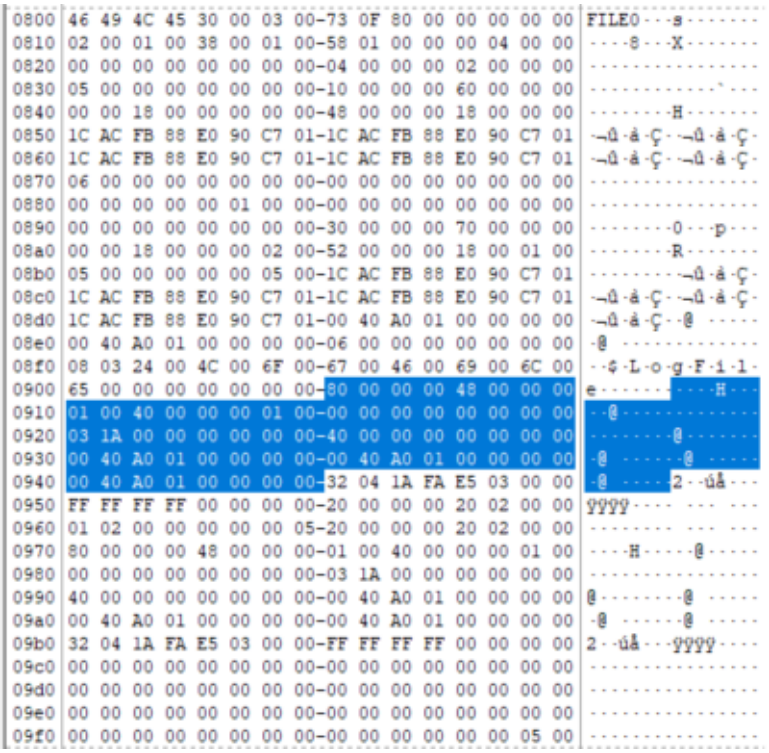
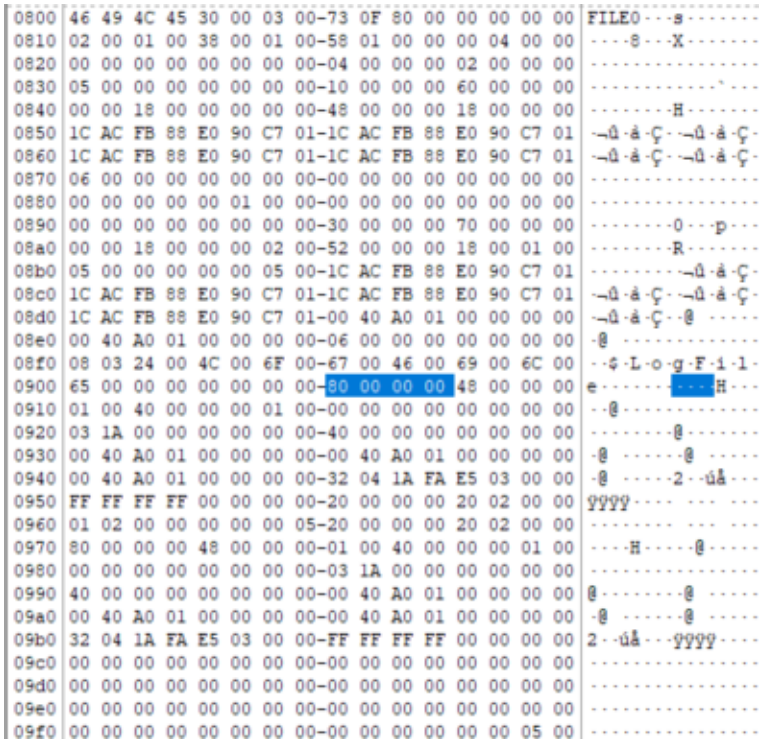
W2 Practical (in Depth)

> Evidence Image can be found in Politemall.
> Question 1 - 5 is straightforward so not covering.
> Below is covering from Point 14 in Word Doc.



Locating $DATA Attribute Header

The highlighted blue section starts at line `0900`, 9th byte:



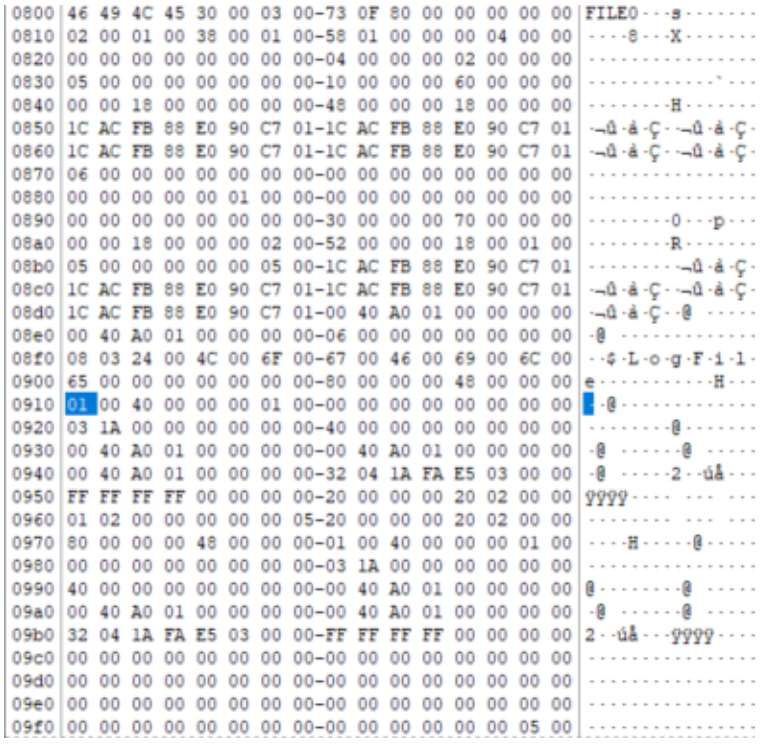
`80 00 00 00` → **Attribute Type Code**: `$DATA` (`0x00000080`).



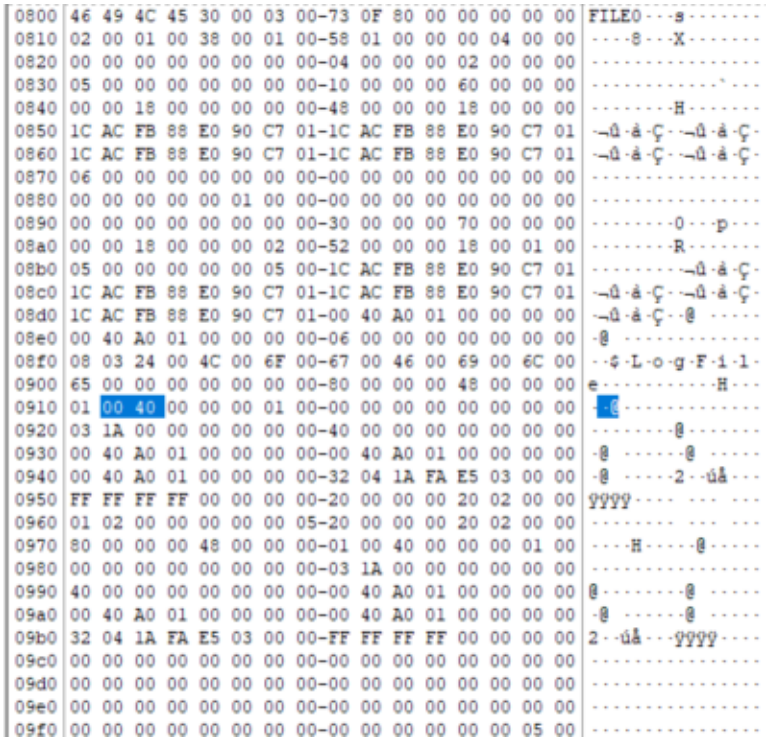
`48 00 00 00` → **Attribute Length**: `0x00000048` = **72 bytes** total for this attribute structure

Non-Resident Flag & Header Structure

Looking at line `0910`:

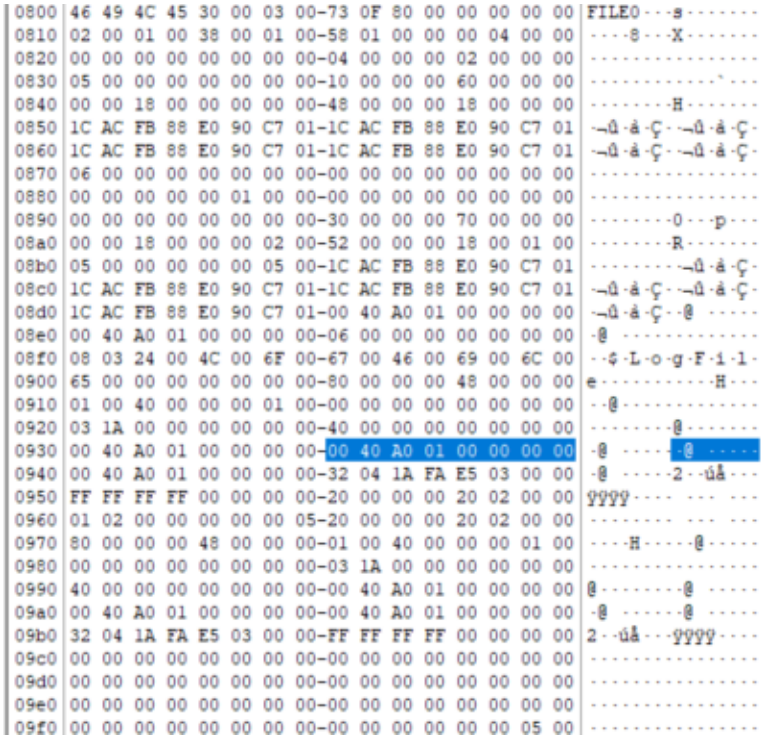


`01` (at offset `0x08` from `80`) → **Non-Resident Flag**: `01` means the file's data is stored in clusters on disk outside the `$MFT`.

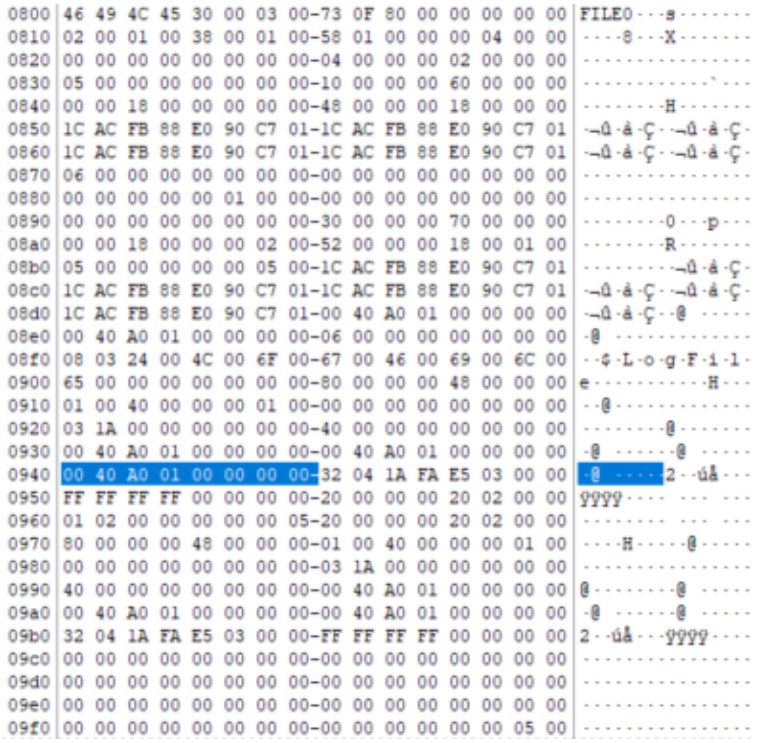


`00 40` (at offset `0x20` from `80`, line `0920`) → **Runlist Offset**: `0x0040` = **64 bytes** from the start of the `80` byte. This points to where the Data Run begins

File Size



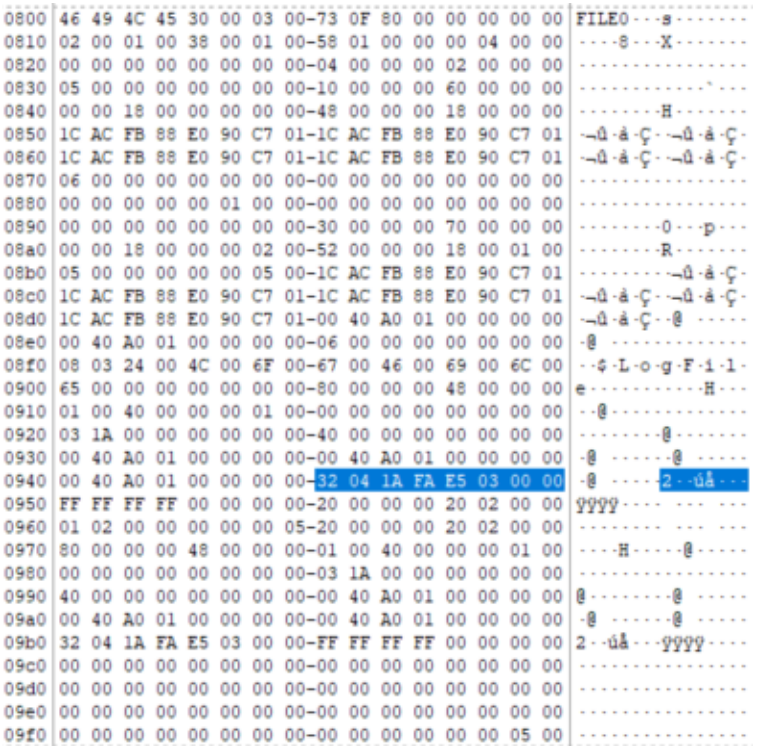
Allocated Size (Offsets `0x28` – `0x2F`): `00 40 A0 01 00 00 00 00` → `0x01A04000` = **27,279,360 bytes** (26 MB).



Actual File Size (Offsets `0x30` – `0x37`): `00 40 A0 01 00 00 00 00` → `0x01A04000` = 27,279,360 bytes

Reading the Data Run (Runlist)

At the very end of the highlighted attribute (line `0940`, starting at the 9th byte):



Header Byte (32)

Low Nibble (2)

- The **Run Length** (number of clusters) uses **2 bytes**.

High Nibble (3)

- The **Starting Cluster Offset** (LCN) uses **3 bytes.**

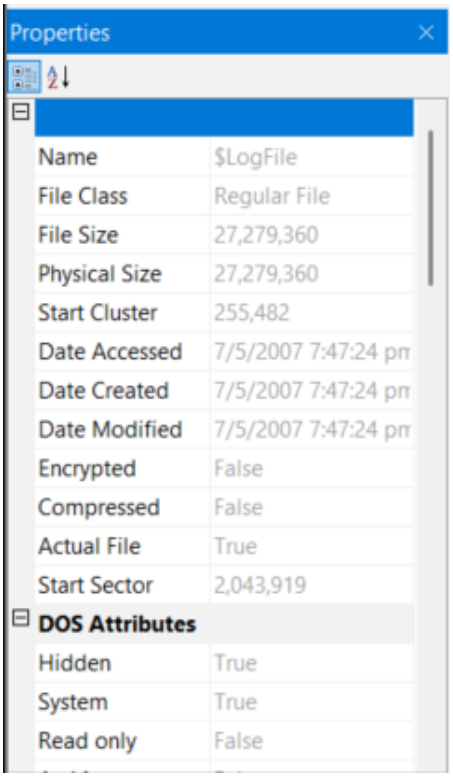
Run Length (Size in Cluster)

- Read next **2 bytes**: `04 1A`
- Reverse (Little-Endian): `0x1A04`
- **Decimal Conversion**: `6,660 clusters`

Starting Cluster Number (LCN):

- Read next **3 bytes**: `FA E5 03`
- Reverse (Little-Endian): `0x03E5FA`
- **Decimal Conversion**: `Cluster 255,482`

18) Compare



FILE SIZE = 27,279,360 ✅ (MATCHES)

START CLUSTER = 255,482 ✅ (MATCHES)

Q6)

Question 6: Describe the steps to compare the 2 sets of values with relevant screenshots of

1. the cluster size and starting cluster number in cluster data runs
2. **Start Cluster**, **Physical Size** displayed in **Properties Pane**.

Answer:

Step 1: Locate and Extract Data Run Values from the Hex Viewing Pane

1. Select `$MFT` in the File List Pane after clicking the `[root]` directory.
2. In the Hex Viewing Pane, search for `$LogFile` (or navigate to record 2 at hex offset `0x08F0`).
3. Locate the `$DATA` attribute header marker `80 00 00 00` (at offset `0x0900`).
4. Move to the end of the non-resident attribute header to locate the **Data Run string**: `32 04 1A FA E5 03 00` (at offset `0x0948`).

5. Parse the Data Run fields:
    - **Header Byte** `32`: Indicates a **2-byte** Run Length and a **3-byte** Run Offset (Starting Cluster).
    - **Run Length (Bytes `04 1A`):** Convert from Little-Endian to hex (`0x1A04`) → `6,660 clusters`.
    - **Starting Cluster / LCN (Bytes `FA E5 03`):** Convert from Little-Endian to hex (`0x03E5FA`) → `Cluster 255,482`.
6. Calculate the **Physical Size in Bytes**:

$$\text{Physical Size} = 6,660 \text{ clusters} \times 4,096 \text{ bytes/cluster} = \mathbf{27,279,360} \text{ bytes}$$

> **Screenshot A to capture:** Take a screenshot of the **Hex Viewing Pane** showing lines `0900` through `0950`, highlighting/marking the `$DATA` header (`80 00 00 00`) and the Data Run sequence (`32 04 1A FA E5 03 00`).

Step 2: Locate Values Displayed in the FTK Imager Properties Pane

1. Click on `$LogFile` in the File List Pane (top right).
2. Look down at the **Properties Pane** located in the bottom-left corner of the FTK Imager window.
3. Observe the displayed metadata values:
    - `First Cluster` / `Start Cluster`: Displays `255482` (or `0x03E5FA`).
    - `Size` / `Physical Size`: Displays `27,279,360 bytes` (or ~26 MB).

> **Screenshot B to capture:** Take a screenshot of the **Properties Pane** (bottom-left) while `$LogFile` is selected, highlighting/marking the **First Cluster** and **Size** fields.

Step 3: Compare and Verify both Sets of Values

Create a comparison table in your report to confirm the values match:

Property	Extracted from Data Run (Hex Pane)	Displayed in Properties Pane	Match Status
Starting Cluster Number	`255,482` (`0x03E5FA` from bytes `FA E5 03`)	`255482`	✅ Match

Cluster Size / Total Clusters	`6,660 clusters` (`0x1A04` from bytes `04 1A`)	`6,660 clusters` (27,279,360 / 4096)	✅ Match
Physical Size (in Bytes)	`27,279,360 bytes` ($6600 * 4,096)	`27,279,360 bytes`	✅ Match

Conclusion

The Starting Cluster (`255,482`) and the calculated Physical Size (`27,279,360 bytes`) parsed from the raw 32-bit Data Run descriptor inside the `$MFT` record directly match the **First Cluster** and **Size** entries auto-parsed by FTK Imager in the Properties Pane.

Question 7

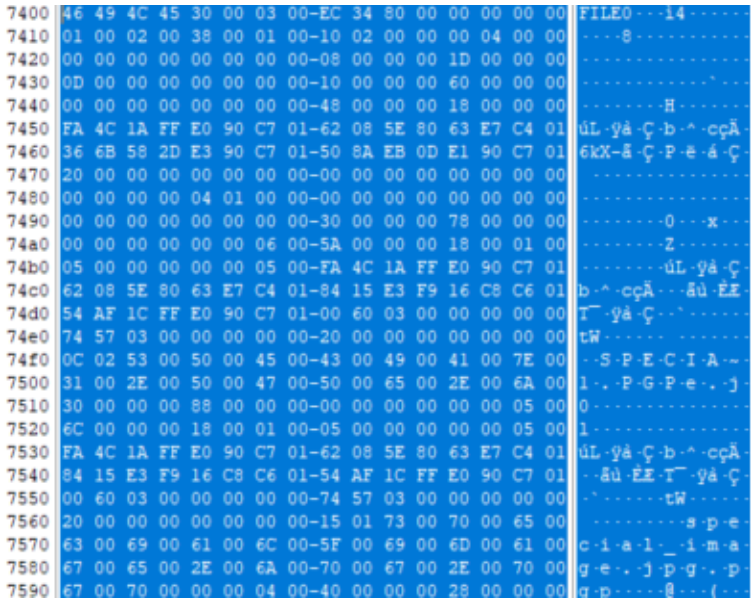
Complete and record the steps with relevant screenshots on the following tasks

a. Locate the File Record of the file **jpg.pgp** in the $MFT file

b. Locate the first cluster data runs

c. Examine and determine the cluster size (in number of clusters) and starting cluster number for each cluster data runs used by the file **special_image.jpg.pgp**

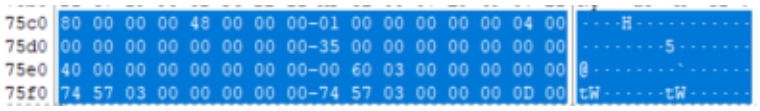
a) Locate the File Record of `special_image.jpg.pgp`



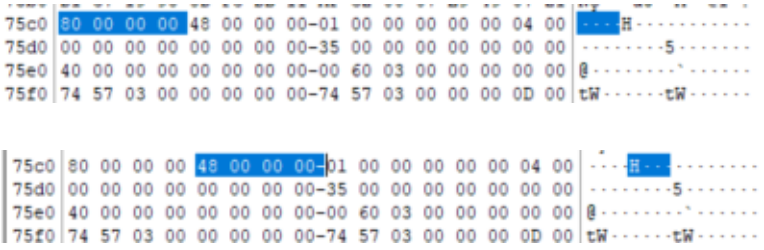
1. Select `[root]` in the Evidence Tree, then select `$MFT` in the File List Pane.
2. Open the Find dialog in the Hex Viewing Pane (`Ctrl + F`) and search for `special_image.jpg.pgp`.
3. The search lands in the file record at offset `0x7400`, where the UTF-16 Unicode string for `s.p.e.c.i.a.l._.i.m.a.g.e...j.p.g...p.g.p` is displayed across rows `7560` **to** `7590`. **Screenshot A to capture:** Highlight row `7400` (the `FILE0` record header) through row `7590` in FTK Imager's Hex Viewing Pane to show the file record boundary and filename attribute.

b) Locate the `$DATA` Attribute Marker (`80 00 00 00`)

Following the `$FILE_NAME` attribute, locate the `$DATA` **attribute header** marked by `80 00 00 00`:



Look at row `75c0`, bytes 0 to 3:



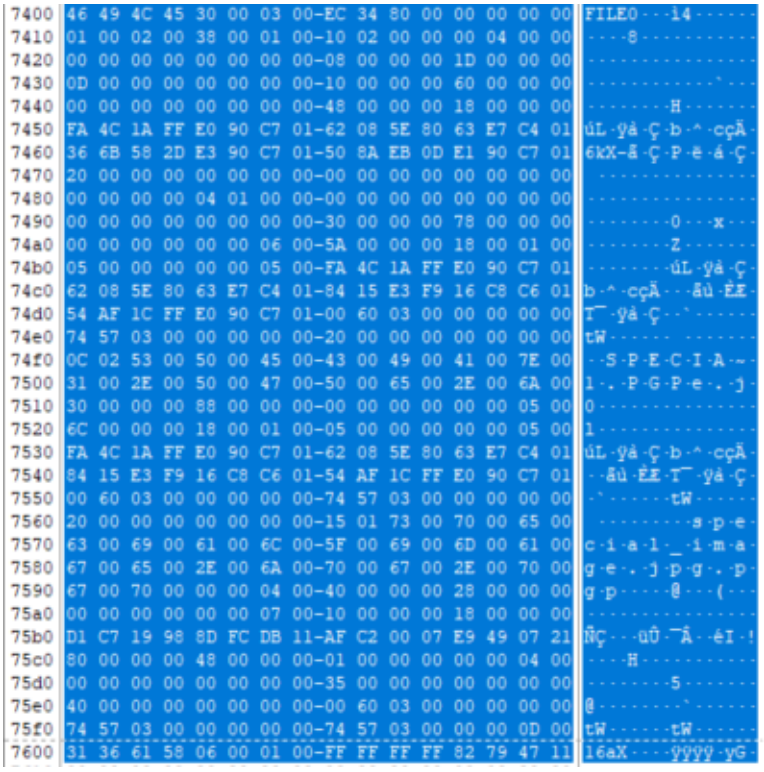

```
75c0: 80 00 00 00 48 00 00 00 ...
      |---------| |---------|
     $DATA Type   Attribute Size (0x48 = 72 bytes)
```

At offset `0x08` from `80` (row `75c0`, byte 8), the byte `01` confirms this `$DATA` stream is **non-resident**.

c) Locate and Parse the First Data Run

Scan to the end of the non-resident `$DATA` header structure to find the Data Run descriptor:

Look at row `7600`, bytes 0 to 4:



- *(Note: Data run bytes start at offset `0x2107E9...` at the end of the 72-byte attribute structure)*

Looking closely at the run bytes `21 07 E9 49`:

Header Byte (`31`):

- **Low Nibble (`1`):** Run Length (Cluster Size) uses **1 byte**.
- **High Nibble (`3`):** Starting Cluster Offset (LCN) uses **3 bytes**.

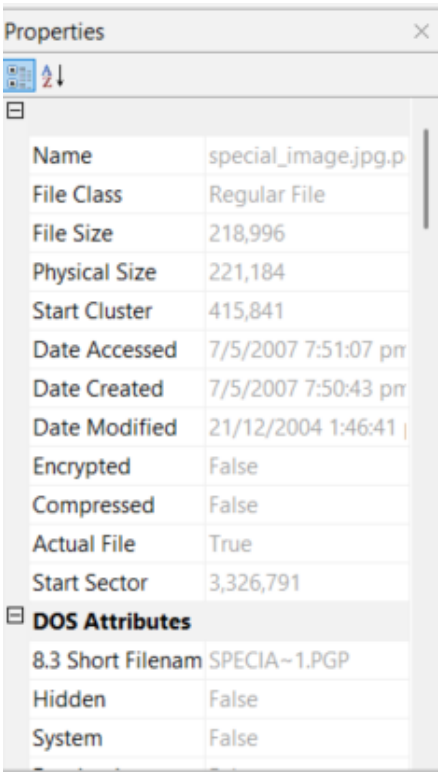
Cluster Size (Run Length - 1 Byte):

- **Byte Value:** `36`
- **Hex Value:** `0x36`
- **Decimal Calculation:** $3 \times 16^1 + 6 = \mathbf{54}$ clusters

Starting Cluster (LCN Offset - 3 Bytes):

- **Raw Bytes:** `61 58 06`
- **Little-Endian Ordering (Reversed):** `0x065861`
- **Decimal Calculation:** $(6 \times 16^4) + (5 \times 16^3) + (8 \times 16^2) + (6 \times 16^1) + 1 = 393,216 + 20,480 + 2,048 + 96 + 1 = \mathbf{415,841}$

Compare



Start Cluster: 415, 841 (MATCHES) ✅